MILITARY SPECIFICATION

FOR

CONNECTOR, ELECTRICAL, POWER

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the performance, design, and test requirements for electrical power connectors associated with the Electrical Power Plant II (EPPII) and Electrical Power Unit (EPU) for the Patriot Missile System.

1.2 Classification. Electrical connectors shall be of the following styles:

1.2.1 Styles.

a. Pin contacts (see 3.1.1)
b. Socket contacts (see 3.1.1)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	- Plating, Cadmium (Electrodeposited).
PPP-B-566	- Box, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock-corner.
PPP-B-636	- Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base.
MIL-L-23699	- Lubricating Oil, Aircraft Turbine Engine.
MIL-C-26074	- Coatings, Electroless Nickel, Requirement for.
MIL-C-55330	- Connectors, Electrical and Fiber Optics, Packaging of.

STANDARDS

MILITARY

MS3197	- Gage Pin for Socket Contact Engagement Test.
MIL-STD-105	- Sampling Procedures and Tables for Inspection of Attributes.
MIL-STD-202	- Test Methods for Electronic and Electrical Components.
MIL-STD-889	- Dissmilar Metals.
MIL-STD-1344	- Test Methods for Electrical Connectors.
MIL-STD-45662	- Calibration System Requirement.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

TA13226E1410	- Connector, Electrical.
TA13226E1440	- Connector, Electrical.
13222E8995	- Cable, Power.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item Description. The electrical power connectors described in this specification are designed to carry 208 VAC 400 Hz 3-phase power and control signals for the EPPII and EPU.

3.1.1 Connectors. When specified (see 6.2), the following connectors shall be furnished:

a. TA13226E1440 for pin-type connector.
b. TA13226E1410 for socket-type connector.

3.2 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the connector. Any data (e.g., shop drawings, layouts, flowsheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be available, upon request, for inspection by the contracting officer or his designated representative.

3.3 First article. The contractor shall furnish one or more connectors as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove prior to starting that his production methods will produce connectors that comply with requirements of this specification. Examination and tests shall be as specified in section 4. Unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examinations and tests, as specified (see 6.2).

3.4 Material. Material shall be as specified herein and on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.4.1 Material deterioration and control. The connectors shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the connector may be exposed.

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3.4.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.5 Performance requirements. The electrical connectors shall comply with the following requirements:

3.5.1 Contact resistance. When connectors are tested as specified in 4.5.3 the resistance of mated pin and socket contacts shall be such that the potential drop at the test current specified in table I shall be not greater than the values specified.

TABLE I. Contact resistance.

Contact	Test curret (amperes)	Potential drop (millivolts, max)	
		Initial	After corrosion
16	20	25	35
1/0	200	10	20

3.5.2 Dielectric withstanding voltage. When connectors are tested as specified in 4.5.4, connectors shall be capable of withstanding 2000 volt AC rms, 60Hz.

3.5.3 Thermal shock. When connectors are tested as specified in 4.5.5 the connectors shall show no evidence of physical damage (see 6.5).

3.5.4 Contact retention. When contacts are tested as specified in 4.5.6 they shall be capable of withstanding the axial loads shown in table II and remain seated in the insert.

TABLE II. Contact retention loads.

Contact size	Axial load (lbs, min)
16 ----------	10
1/0 ----------	35

3.5.5 Insert retention. When insert assemblies are tested as specified in 4.5.7 they shall retain their normal position in the connector shell for a minimum of 5 seconds at the 45 pounds force per square inch (lb_f/in^2) gage pressure.

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3.5.6 Insulation resistance. When connectors are tested as specified in 4.5.8, the initial insulation resistance shall be not less than 5,000 megohms.

3.5.7 Moisture resistance. When connectors are tested as specified in 4.5.9, they shall meet the following requirements:

Insulation resistance:
During the 10th cycle ---------- 10 megohms minimum
After 24-hour drying period ---------- 1,000 megohms minimum
Dielectric withstanding voltage ---------- As specified in 3.5.2

3.5.8 Durability. When connectors are tested as specified in 4.5.10, there shall be no evidence of damage. Flaking of the plating is considered a failure.

3.5.9 Cable pull-out. When connectors are tested as specified in 4.5.11, test cables (see 4.2.3) shall not pull-out when a load of 75 pounds is applied, nor shall the slippage exceed 1/8 inch.

3.5.10 Vibration. When the wired connector assembly is tested as specified in 4.5.12, there shall be no cracking, breaking, or loosening of parts, nor shall there be a loss of electrical continuity in excess of 10 microseconds.

3.5.11 Water immersion. When tested in accordance with 4.5.13, the connector shall show no leakage. There shall be no evidence of water leakage at the connector interface of mated connectors. At the end of 4 hours while the mated plugs are still immersed, the insulation resistance shall be 100 megohms minimum.

3.5.12 Fluid immersion. When connectors are tested as specified in 4.5.14, they shall mate properly with their counterpart connectors and withstand one-half the applicable voltage stipulated in 3.5.2.

3.5.13 Contact engagement and separation forces. When socket contacts are tested as specified in 4.5.15, they shall comply with the contact engagement and separation forces in table III.

TABLE III. Contact engagement and separation forces.

Contact size	Force (pounds)	
	maximum	minimum
16 ----------	3.0	0.25
1/0 ----------	20.0	2.00

3.5.14 Shell to shell conductivity. With the connector mated with counterpart receptacle, the voltage drop measured from the rear end of the

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connector shell to a point on the front of the receptacle flange shall not exceed 5.0 millivolts DC for a current of 1.0 ±0.1 ampere DC.

3.6 Marking. Each connector, shall be legibly and permanently marked as specified on the drawing referenced herein.

3.7 Workmanship. The connector shall be processed in such a manner as to be uniform in quality and shall be free from defects that will affect life, serviceability, or appearance. There shall be no evidence of poor molding or fabricating; damaged contacts; peeling or chipping of the plating and finish; excessive flash which would indicate improper molding; nicks, and burrs to metal surfaces. External surfaces shall be free of sharp edges and corners except where sharp edges or corners are required or where they are not detrimental to safety.

3.7.1 Plating. The connector shall be nickel plated 0.001 inch in thickness (minimum) in accordance with MIL-C-26074, class 4, grade A. A conductive coating of cadmium plating 0.0001 inch minimum, thickness in accordance with QQ-P-416, type II, color olive drab, shall be plated over the nickel coating. Workmanship for the nickel plating shall be in accordance with MIL-C-26074.

3.7.2 Rubber parts. All rubber parts shall be free of flashing, nicks, porosity, and molding or manufacturing flaws that would make the part unsuitable for its intended use.

3.7.3 Rivets. Rivets shall be installed in accordance with the drawings referenced herein.

3.7.4 Threads. The connector threads shall be free from burrs, nicks, and other defects that would cause binding or would not allow threaded parts to be mated smoothly.

3.8 Government furnished equipment. When specified (see 6.2) the Government will furnish the following equipment for the conduct of test specified herein:

a. Receptacle, MIS-20045/2 - 019 (socket type)
b. Receptacle, MIS-20045/2 - 020 (pin type)
c. Cable, Power, Drawing 13222E8995

4. Quality assurance provisions.

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements, specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is resposible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any test or examination by the first article model shall be cause for disassembly, in the presence of a Government representative, of the first article model to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first article model. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.2 Classification of inspections. The connectors shall be inspected as specified herein. Inspections shall be classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)
c. Inspection of packaging (see 4.6)

4.2.1 Inspection conditions. Unless otherwise specified, inspections required by this specification shall be made under any combinations of conditions within the following ranges. Any specified condition shall not affect the other two ambient ranges.

a. Temperature 15° C to 35° C.
b. Relative humidity 30 to 80 percent.
c. Barometric pressure 650 to 850 mm of mercury.

4.2.2 Test equipment calibration. Calibration of test equipment shall be in accordance with MIL-STD-45662.

4.2.3 Test cables. For the purpose of these tests, the connectors shall be wired with cable conforming to drawing 13222E8995 (see 3.8).

4.3 First article inspection.

4.3.1 Examination. The first article connector(s) shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection and for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. Upon successful completion of the examination specified in 4.3.1, the first article connector(s) shall be subjected to the tests marked "X" in column 1 of table V. Failure of any test shall be cause for rejection and for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for tests shall be in accordance with MIL-STD-105, inspection level II.

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4.4.2 Individual. Each connector shall be examined as specified in column 2 of table V. Presence of one or more defects shall be cause for rejection.

4.4.3 Samples. Connectors shall be subjected to the tests marked "X" in column 3 of table V. The acceptable quality level (AQL) shall be 2.5 percent defective. Presence of one or more defects shall be cause for rejection.

4.5 Examination.

4.5.1 Visual and mechanical inspection. The connectors shall be examined for the following defects in table IV.

TABLE IV. Classification of defects.

Major	Defects	Requirements
101.	Plating not as specified.	3.7.1
102.	Rubber parts not as specified.	3.7.2
103.	Contact insert not installed correctly.	3.1.1
104.	Contact spring not installed correctly.	3.1.1
105.	Connector contains scratches, bubbles in plating, or other workmanship defects.	3.7
106.	Connector not identified as specified.	3.6
107.	Connector assembly does not contain all parts.	3.1.1
108.	Connector threads not free from burrs, nicks or other defects that would affect function.	3.7.4
109.	Connector assembly not free of sharp edges or burrs.	3.7
110.	Rivets in protective caps and retaining straps not installed correctly.	3.7.3
111.	Materials not as specified.	3.4
112.	Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.	3.4.1
113.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.	3.4.2
114.	Workmanship not as specified.	3.7

4.5.2 Quality conformance inspection. The connector shall be tested for the requirements identified in table V in accordance with the test procedures contained herein.

4.5.3 Contact resistance. Each pair of mated pin and socket contacts shall be tested after only one mating of the contacts, in accordance with method 3004 of MIL-STD-1344. The following details and exceptions shall apply:

a. Wire size and type - See 3.8.

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b. Test current - See table I.
c. Test sample preparation - In addition to preparation in method 3004, the unassembled contacts shall be mated to the minimum depth as determined when the contacts are installed in the appropriate connectors. The potential drop shall be measured at the extreme terminal ends of the contacts. See figure 1 for wiring diagram.
d. Millivolt drop requirements - See table I.

Noncompliance with 3.5.1 shall be cause for rejection.

4.5.4 Dielectric withstanding voltage. Unmated connectors shall be tested in accordance with method 3001 of MIL-STD-1344. The following details and exceptions shall apply:

a. Test voltage - 2000 volt ac, rms, 60 Hz.
b. Points of application of test voltage - Between the two closest contacts and between the shell and the contacts closest to the shell.
c. Application of test voltage - The test voltages shall be applied gradually at the rate of approximately 500 volts per second until the specified voltage is reached.

Noncompliance with 3.5.2 shall be cause for rejection.

4.5.5 Thermal shock. Unmated connectors shall be tested in accordance with method 1003, test condition A, of MIL-STD-1344, except that the low temperature shall be $-55^{\circ}\ ^{+0^{\circ}C.}_{-3}$. Noncompliance with 3.5.3 shall be cause for rejection.

TABLE V. Quality conformance inspection.

First Article	Quality Conformance Inspection	Quality Conformance Sample	Examination or Test	Requirement Paragraph	Test
1	2	3	4	5	6
X	X		Visual & mechanical.	3.1, 3.4, 3.6, 3.7	4.5.1
X		X	Contact resistance.	3.5.1	4.5.3
X			Dielectric withstanding voltage	3.5.2	4.5.4
X		X	Cable pull-out.	3.5.9	4.5.11
X			Contact retention	3.5.4	4.5.6

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TABLE V. (Continued).

First Article	Quality Conformance Inspection	Quality Conformance Sample	Examination or Test	Requirement Paragraph	Test
1	2	3	4	5	6
X		X	Insert retention.	3.5.5	4.5.7
X		X	Insulation resistance.	3.5.6	4.5.8
X(1)			Contact engagement & separation forces. .	3.5.13	4.5.15
X		X	Shell to shell Conductivity.	3.5.14	4.5.16
X			Thermal shock.	3.5.3	4.5.5
X			Moisture resistance.	3.5.7	4.5.9
X			Vibration.	3.5.10	4.5.12
x		X	Water immersion.	3.5.11	4.5.13
X			Fluid immersion.	3.5.12	4.5.14
X			Durability	3.5.13	4.5.10

(1) Socket type contacts only.

4.5.6 Contact retention. Individual contacts shall be subjected to the axial loads specified in table II. The load shall be applied, first in one direction, then the other, on individual contacts with all other contacts in place and the insert in the shell, uniformly at a rate of approximately 1 pound per second. Failure of the contacts to remain seated in the insert at the specified axial load (see 3.5.4) shall be cause for rejection.

4.5.7 Insert retention. Inserts shall be subjected to the axial load specified in 3.5.5, first in one direction, then the other. For sample tests, rubber plugs shall be used to seal the contact holes in the inserts. Loading shall be accomplished by applying air pressure alternately to each face of the insert. The pressure shall be increased gradually at a rate of approximately 10 lb_f/in^2 per second until the specified pressure in 3.5.5 is reached. Failure of the insert to remain seated for the specified axial load for 5 seconds (minimum) shall be cause for rejection.

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4.5.8 Insulation resistance. Mated connectors shall be tested as specified in method 3003 of MIL-STD-1344. Points of application of test voltage shall be between all adjacent pairs of contacts. Nonconformance to 3.5.6 shall be cause for rejection.

4.5.9 Moisture resistance. Connectors shall be completely wired and assembled. Cable used shall be as specified (see 3.8). The mated connectors shall be subjected to 10 cycles of exposure in accordance with method 106 of MIL-STD-202, with the exceptions and details listed below:

a. Connectors shall be mounted in a horizontal position with the wires or cables descending into the backshells. There shall be no drip loops in the wires or cables.
b. Wires and cables shall be brought out of the chamber through vapor-tight seals or ends of the cables or wires effectively sealed.
c. There shall be no wire splices in the chamber.
d. Delete steps 7a and 7b.
e. During steps 1 and 6 inclusive, a polarizing potential of 100 volts dc shall be applied between alternate contacts connected together electrically and the remaining contacts and shell connected together electrically. The polarity of the voltage applied to the shell shall be negative.
f. After a minimum of 3 hours of step 7 of the tenth cycle, while at the high humidity condition, the connectors shall be subjected to the insulation resistance test of 4.5.8.
g. Following these measurements, the connectors shall be maintained at test conditions of 4.2.1 for 24 hours maximum. At the conclusion of this recovery period, insulation resistance and dielectric withstanding voltage shall be tested as specified in 4.5.8 and 4.5.4, respectively.

Nonconformance to 3.5.7 or damage to the connector shall be cause for rejection.

4.5.10 Durability. The complete connector, less coupling ring, shall be mated and unmated 500 times at a rate not exceeding 100 cycles per hour. Mating and unmating shall be accomplished in a manner similar to that which the connector will be subjected in service. Nonconformance to 3.5.8 shall be cause for rejection.

4.5.11 Cable pull-out. Cables assembled in the connector, but with the conductors not attached to the contacts, shall be subjected to the axial tensile load pull-out force specified in 3.5.9. The amount of slippage shall be measured between the cable and the connector adapter. The specified load shall be applied for a minimum of 1 hour. Nonconformance to 3.5.9 shall be cause for rejection.

4.5.12 Vibration. Connectors shall be tested in accordance with method 2005 of MIL-STD-1344. The following details and exceptions shall apply:

a. Test condition number - III
b. The connector shall be engaged with the mounted receptacles and held by normal locking means only. No safety wire shall be used.

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c. The vibration of the receptacle and fixture shall be monitored by suitable sensor at a point on the fixture near a receptacle support or on the receptacle itself.
d. All contacts shall be wired in a series circuit with at least 100 milli-amperes of current flowing through the series circuit during vibration.
e. The cable shall be clamped to nonvibrating points at least 8 inches from the rear of the connectors. The clamping length may be selected or changed to avoid resonance of the cable or wire.

Nonconformance to 3.5.10 shall be cause for rejection.

4.5.13 Water immersion (mated). All connectors shall be completely wired and assembled. Cable used shall be in accordance with 3.8. Connectors assembled to cables shall be tested in a chamber under pressure by submerging in conductive water. The water conductivity shall be assured by the addition of at least 5 percent sodium chloride by weight. The cable ends shall be sealed or extended outside the chamber. The tests shall be performed after each of the following conditionings:

a. Room temperature (see 4.2.1).
b. After a minimum of 4 hours at 80° $^{+3°}_{-0°}$ C and return to room temperature.

4.5.13.1 Mated. The chamber shall be at a 1 standard atmosphere (atm) pressure differential for four hours minimum. While the samples are still submerged and after the chamber has returned to normal pressure, insulation resistance shall be measured as specified in 4.5.8. Nonconformance to 3.5.6 shall be cause for rejection.

4.5.14 Fluid immersion. Unmated connectors shall be immersed fully in the fluids specified in table VI for 20 hours minimum. Upon removal from the fluid, connectors shall be maintained in free air at room temperature for 1 hour maximum. Dielectric withstanding voltage shall then be tested as specified in 4.5.4, except the magnitude of test voltage shall meet the requirements of 3.5.12. The connector shall be mated with its mating receptacle to demonstrate that there is no damage to the inserts that would cause interference in mating the connector. Failure to mate with mating receptacle or comply with 3.5.12 shall be cause for rejection.

TABLE VI. Fluids for fluid immersion.

Fluids	Specification
Aviation hydraulic fluid (petroleum base).	MIL-L-5606
Aircraft lubrication oil.	MIL-L-23699

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4.5.15 Contact engagement and separation forces. Sockets shall be mounted in a suitable position or fixture for applying gradually increasing loads for the engagement and separation of test pins. Half of the quantity of socket contacts used in this test shall be installed in inserts and the remaining socket contacts shall be out of inserts. Depth of engagement shall conform to that encountered in service. The test pins shall be in accordance with MS3197. The minimum diameter test pin shall be inserted and removed from the socket and the separation force measured during removal. The maximum diameter test pin shall be inserted and removed from the socket and the engagement force measured during insertion. Nonconformance to 3.5.13 shall be cause for rejection.

4.5.16 Shell to shell conductivity. With the connector mated to the counterpart receptacle, conduct the test specified in method 3007 of MIL-STD-1344. Nonconformance to 3.5.14 shall be cause for rejection.

4.6 Inspection of packaging.

4.6.1 First article packaging inspection. The first article pack shall be examined for defects listed in 4.6.2.3. The presence of one or more defects shall be cause for rejection of the first article pack.

4.6.2 Quality conformance inspection of packaging.

4.6.2.1 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.2.3 Examination. Samples selected in accordance with 4.6.2.2 shall be examined for the following defects. The AQL shall be 2.5 percent defective.

No.	Defect	Levels		
		A	B	C
115	Preparation guidelines not followed.	5.2	5.2	5.2
116.	Cleaning process not as specified.	5.3.1.1		5.3.2
117.	Connector not properly dried.	5.3.1.2		
118.	Contact preservative applied connector.	5.3.1.3		5.3.2
119.	Unit container not as specified.	5.3.1.4		5.3.2
120.	Intermediate container not as specified.	5.3.1.5		5.3.2

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No.	Defect	Levels		
		A	B	C
121.	Gross weight of intermediate container exceeds weight limitations.	5.3.1.5		
122.	Closure of intermediate container not as specified.	5.3.1.5		
123.	Shipping container not as specified.	5.4.1	5.4.2	5.4.3
124.	Closure and strapping of shipping container not as specified.	5.4.1	5.4.2	
125.	Gross weight of shipping container exceeds the maximum specified weight.	5.4.1	5.4.2	
126.	Marking not as specified.	5.5.1	5.5.1	5.5.2

5. Packaging.

5.1 First article packaging. Unless otherwise specified (see 6.2), the contractor shall furnish a first article pack for examination within the time frame specified (see 6.2), to prove prior to the start of production packaging, that applied preservation, packaging and marking comply with the requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may utilize either first article models or production models. When first article models are utilized, and the Government requests a comparison between a first article model and a production model, any preservation and packing shall be removed by the contractor at no expense to the Government.

5.2 General. Each connector, as applicable, shall be prepared for shipment in accordance with the requirements of the general guidelines of MIL-C-55330.

5.3 Preservation. Preservation shall be level A, or commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Cleaning. Each complete connector shall be cleaned in accordance with MIL-P-116, process C-1.

5.3.1.2 Drying. Each complete connector, cleaned as specified in 5.3.1.1, shall be dried in accordance with MIL-P-116.

5.3.1.3 Preservatives. Contact preservatives shall not be used.

5.3.1.4 Unit container. Each complete connector, cleaned and dried as specified in 5.3.1.1 and 5.3.1.2 respectively, shall be placed in a water-vapor-proof bag in accordance with method 1A-8 of MIL-P-116. The connector shall then

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be placed in a snug-fitting fiberboard box. the snug-fitting box shall comply with PPP-B-566 or PPP-B-636, or shall be the contractor's standard commercial snug-fitting fiberboard box. Box closure shall be with reinforced tape.

5.3.1.5 Intermediate container. Like connectors, in unit containers as specified in 5.3.1.4, shall be placed in a snug-fitting fiberboard box conforming to PPP-B-636, class-weather resistant. The quantity of connectors per box shall not result in a gross weight which exceeds the weight limitations of the box specification. Box closure shall be method V of the appendix thereto.

5.3.2 Commercial. Each connector shall be preserved in accordance with the requirements of ASTM D 3951. Contact preservatives shall not be used.

5.4 Packing. Packing shall be level A, level B or commercial as specified (see 6.2).

5.4.1 Level A. Like connectors, preserved as specified in 5.3, shall be packed in a snug-fitting box conforming to PPP-B-601, overseas type, style optional or PPP-B-621, class 2, style optional. Box closure and strapping shall be as specified in the applicable box specification or the appendix thereto, except that the strapping shall be flat and the finsih B. The gross weight of the box shall not exceed 750 pounds.

5.4.2 Level B. Like connectors, preserved as specified in 5.3, shall be packed as specified in 5.4.1 for level A except that the boxes shall be domestic style or class and the strapping finish shall be A or B.

5.4.3 Commercial. Like connectors, preserved as specified in 5.3, shall be packed in accordance with ASTM D 3951.

5.5 Marking.

5.5.1 Levels A and B. In addition to any special or identification marking required by the contract or purchase order, each unit, intermediate and exterior container shall be marked in accordance with MIL-STD-129.

5.5.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Additionally, each exterior container shall be marked with the cube and gross weight.

6. Notes.

6.1 Intended use. The connectors described in this specification are intended for use on the EPPII and EPU of the PATRIOT Missile System.

6.2 Ordering data. Procurement documents should specify the following:

a. Title and date of this specification.
b. Type of connector required (see 3.1.1).
c. Number of first article samples and time of delivery (see 3.3).

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d. When the Government will conduct any or all first article examinations and test (see 3.3).
e. Government furnished equipment to be supplied (see 3.8).
f. When a first article pack is not required (see 5.1).
g. Time frame required for submission of first article packs when required (see 5.1).
h. Degree of preservation and packing (see 5.3 and 5.4).
i. Any special marking (see 5.5.1).

6.3 First article. When a first article inspection is required, the items should be a preproduction connector. The first article should consist of 1 or more units. The contracting officer should include specific instructions in acquistion documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the document's first article. Any changes or deviation of production connectors from the approved preproduction connector will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish connectors conforming to this specification.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack during production will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack and mark the connectors in accordance with this specification.

6.5 Damage. Damage is defined but not limited to cracks or bubbles in plating, swelling of rubber parts, loosening of metal parts or other physical changes in the connector that would prevent the connector from performing its intended function.

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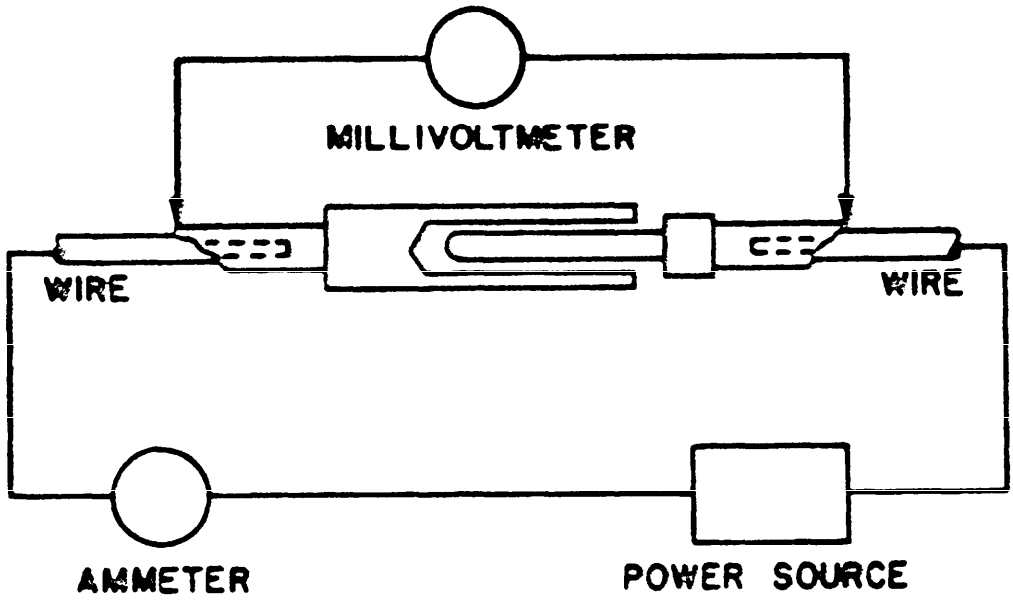


FIGURE 1. Wiring diagram for contact resistance test.

X-4314

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-53052(ME)	Connector, Electrical, Power

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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